Beyond Grammar: Detecting and Classifying Meaning-Related Errors in L2 Writing with Large Language Models

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https://github.com/norikotaka/l2-meaning-error-analysis.git

Abstract

In L2 writing, many errors are not strictly grammatical but relate to unnatural expression, weak structure, or unclear meaning. While prior NLP research has mainly focused on grammatical error correction, less attention has been given to meaning-related errors that affect communicative quality. This study addresses this gap by proposing a span-level annotation framework with three categories: expression, discourse, and meaning errors. Using learner essays from the ICNALE corpus, an annotated dataset is constructed to examine whether a large language model can detect and classify these error types. Experiments are conducted using the Qwen3-8B model, comparing zero-shot and few-shot prompting conditions. Performance is analyzed across label complexity, L1 groups, and proficiency levels. An additional L1 identification task is included to assess whether the data captures L1-specific Patterns. The results show that while error detection is relatively easy, error type classification is substantially more challenging. The model performs best on discourse errors and struggles most with meaning errors. Few-shot prompting improves precision but reduces recall, and performance varies depending on label overlap, L1 background, and proficiency level. These findings highlight the difficulty of modeling deeper aspects of language use and suggest the need for more fine-grained annotation and modeling approaches for meaning-related errors.

1. Introduction

In L2 writing, not all errors are equally visible or equally important. While grammatical errors have traditionally received the most attention, many sentences that are grammatically acceptable still sound unnatural, poorly structured, or difficult to interpret. These issues often arise from inappropriate lexical choices, weak discourse organization, or unclear meaning, rather than from surface-level grammatical mistakes. As a result, focusing only on grammar provides an incomplete picture of learner writing quality.

In both L2 research and NLP, most existing work has concentrated on surface-level errors. Large-scale datasets such as the NUS Corpus of Learner English and the BEA-2019 shared task data have been designed primarily for grammatical error correction, emphasizing token-level edits and form-based evaluation (Dahlmeier et al., 2013; Bryant et al., 2019). These resources have enabled strong performance in detecting and correcting grammatical mistakes, but they offer limited support for analyzing errors that affect communicative adequacy.

From a theoretical perspective, language use involves multiple dimensions. Larsen-Freeman (2001) conceptualizes language as an integration of form, meaning, and use, suggesting that breakdowns can occur even when surface form appears acceptable. In parallel, research in L2 pragmatics has demonstrated that communicative appropriateness and interpretability are measurable aspects of language competence (Taguchi, 2015; Ren, 2022). However, these perspectives have not been widely translated into annotation schemes for naturally occurring learner writing.

This study addresses this gap by proposing a span-level annotation framework for meaning-related errors in L2 writing. I define three categories: expression errors, which involve unnatural lexical or phrasal choices; discourse errors, which involve problems in structure, flow, or coherence; and meaning errors, which involve unclear or difficult-to-interpret content. Using this framework, I construct a dataset based on learner essays and investigate whether such errors can be annotated reliably and whether computational models can detect and distinguish them.

2. Errors in L2 Writing

Research on L2 writing has traditionally emphasized grammatical correctness, largely because grammatical errors are relatively easy to define, annotate, and evaluate. Learner corpora such as NUCLE provide detailed annotations of grammatical errors, and shared tasks such as BEA-2019 have further reinforced the focus on correction-oriented tasks (Dahlmeier et al., 2013; Bryant et al., 2019). While this line of work has been highly successful, it primarily captures surface-level deviations from target language norms.

At the same time, a lot of research suggests that other types of errors play an equally important role in writing quality. Learner corpus studies have shown that advanced learners frequently produce unnatural collocations and phraseological patterns, even when their grammar is relatively accurate (Granger, 1998). These lexical issues affect the perceived naturalness of writing and cannot be reduced to simple grammatical mistakes. In addition, L2 writing research has emphasized the importance of discourse-level competence. Effective writing requires coherent organization, appropriate use of connectives, and logical flow across sentences, all of which pose challenges for learners (Hyland, 2002; Hinkel, 2002).

Beyond lexical and discourse issues, errors may also arise at the level of meaning. According to Larsen-Freeman (2001), learners may produce sentences that are grammatically well-formed but semantically unclear or pragmatically inappropriate. Research in L2 pragmatics further supports the idea that meaning and appropriateness are central to language use and can be systematically studied (Taguchi, 2015; Ren, 2022). However, these aspects have rarely been incorporated into annotation frameworks for learner writing.

Taken together, prior work suggests that L2 writing errors should be understood across multiple levels, including lexical choice, discourse organization, and meaning construction. The present study operationalizes these dimensions into three annotation categories. (1) expression, (2) discourse, and (3) meaning to capture different types of communicative breakdown.

3. NLP in L2 Writing

NLP approaches to learner writing have achieved substantial progress, particularly in grammatical error correction. Datasets such as NUCLE and the BEA-2019 shared task have enabled the development of models that can automatically detect and correct surface-level errors with high accuracy (Dahlmeier et al., 2013; Bryant et al., 2019). These tasks are typically formulated as sequence-to-sequence or token-level prediction problems, focusing on edits that transform learner text into well-formed target text.

Despite these advances, most NLP work on learner writing remains centered on surface features. Errors are often defined in terms of token-level changes, and evaluation focuses on how closely model outputs match reference corrections. As a result, errors that affect meaning, discourse, or pragmatic appropriateness are less frequently modeled.

From a learner corpus perspective, annotation design plays a crucial role in determining what aspects of learner language can be analyzed. Jarvis and Paquot (2015) emphasize that the choice of annotation scheme shapes the kinds of patterns that can be observed, while Meurers (2015) highlights the importance of integrating linguistic theory with computational methods. These perspectives suggest that expanding annotation schemes beyond grammar is necessary to capture more complex aspects of learner writing.

However, there is currently a lack of datasets that explicitly annotate meaning-related errors at the span level. This gap motivates the present study.

4. Research Questions

Based on these gaps, this study addresses two research questions:
First, can meaning-related L2 writing errors be reliably annotated by human raters?
Second, can large language models, such as Qwen3, detect and classify these errors?

5. Method

5.1 Dataset

This study uses the ICNALE corpus, which contains argumentative essays written by learners from various Asian L1 backgrounds. All essays were written in response to shared argumentative prompts, and the corpus provides learner metadata, including L1 background and CEFR-linked proficiency level.

I initially selected essays from three L1 groups: Chinese, Japanese, and Korean. For each L1 group, I selected essays from four proficiency levels: A2, B1 low, B1 high, and B2. The original sampling plan included five essays per proficiency level within each L1 group, resulting in 20 essays per L1 group and 60 essays in total. This design was intended to create a balanced sample across both L1 background and proficiency level.

5.2 Error Span Extraction

Candidate error spans are automatically extracted using a GPT-4o model. The model is prompted to identify parts of sentences that may sound unnatural, poorly structured, or unclear. These automatically generated spans serve as the basis for human annotation. Annotators are asked to verify the correctness of span boundaries by marking each span as either “Agree” or “Disagree.” The full prompt used for GPT-4o span extraction is provided in the appendix.

5.3 Annotation Procedure

Annotation was conducted by students enrolled in a language annotation course (total 20 students). Each annotator was assigned essays and asked to label pre-extracted error spans. Each row in the annotation interface corresponds to a single error span, and sentences with multiple issues appear multiple times with different spans.

The annotation task focuses on communicative adequacy rather than surface correctness. Annotators are instructed to identify errors that affect naturalness, structure, or clarity, while ignoring minor grammatical issues that do not interfere with understanding. For each span, annotators assign binary labels indicating whether the span corresponds to expression, discourse, or meaning errors. Multiple labels can be assigned to a single span if multiple issues are present. The detailed annotation instructions are included in the appendix.

5.4 Three Error Categories

Expression errors capture cases where the wording is understandable but sounds unnatural or non-native-like. Discourse errors capture problems in sentence structure, logical flow, or stylistic appropriateness. Meaning errors capture cases where the intended meaning is unclear or difficult to interpret. These categories are designed to reflect different levels of communicative breakdown, drawing on prior work in lexical analysis, discourse studies, and L2 pragmatics.

5.5 Annotation Data and Inter-rater Reliability

5.5.1 Dataset Distribution

For the annotation task, 20 annotators were originally recruited, with each annotator assigned six essays. However, the final annotation pool included 19 annotators. After retaining only essays with at least two assigned annotators, the dataset included 25 essays and 990 annotator-level rows. Because some annotators did not complete all assigned spans or left some fields blank, not every span had two usable annotations. Therefore, the final inter-rater reliability analysis was conducted on 489 unique spans with valid paired annotations from two raters.

Although the original sampling plan was balanced across L1 background and proficiency level, the final reliability dataset was not fully balanced. This imbalance reflects the availability of completed paired annotations rather than the original essay selection design.

Among the final paired spans, 452 were marked as errors and 37 were marked as non-errors. At the label level, Expression was the most frequent label, with 288 positive cases, followed by Discourse with 235 positive cases and Meaning with 179 positive cases. Table 1 summarizes the overall dataset distribution.

Table 1. Overall dataset count

Category	Negative / No error	Positive / Error
Span error	37	452
Expression	201	288
Discourse	254	235
Meaning	310	179

The dataset was not fully balanced across L1 and fluency groups because only spans with valid paired annotations were retained for reliability analysis. Chinese essays contributed the largest number of spans, followed by Japanese and Korean essays. Some L1 and fluency combinations were not represented in the final paired-span dataset. Table 2 shows the number of essays, spans, span-level error counts, and label-level counts by L1 and fluency.

Table 2. Dataset count by L1 and fluency

L1	Fluency	Essays	Spans	No error 0	Error 1	Exp 0	Exp 1	Dis 0	Dis 1	Mean 0	Mean 1
CHN	A2	5	94	4	90	34	60	39	55	62	32
CHN	B1_1	4	86	5	81	27	59	41	45	50	36
CHN	B1_2	3	57	12	45	29	28	44	13	47	10
CHN	B2	2	29	0	29	14	15	19	10	26	3
JPN	A2	2	46	3	43	17	29	19	27	26	20
JPN	B1_2	1	20	0	20	6	14	5	15	9	11
JPN	B2	2	38	5	33	16	22	30	8	28	10
KOR	A2	3	50	2	48	27	23	18	32	25	25
KOR	B1_1	2	48	6	42	31	17	39	9	37	11
KOR	B1_2	1	21	0	21	0	21	0	21	0	21

Note. For span error, 0 indicates that the candidate span was not judged to be an error, while 1 indicates that it was judged to be an error. For the three error labels, 0 indicates that the label was not assigned and 1 indicates that the label was assigned.

Because the annotation scheme allowed overlap among the three error-type labels, each span could receive zero, one, or multiple labels. To examine label complexity, I grouped spans by their label pattern across Expression, Discourse, and Meaning. A pattern of 1 indicates that the label was assigned, while 0 indicates that it was not assigned. (See Table 3)

Table 3. Label pattern distribution

Label pattern	Meaning	Count	Group
0,0,0	No error-type label	99	No label
1,0,0	Expression only	94	Single-label
0,1,0	Discourse only	43	Single-label
0,0,1	Meaning only	25	Single-label
1,1,0	Expression + Discourse	74	Multi-label
1,0,1	Expression + Meaning	36	Multi-label

0,1,1	Discourse + Meaning	34	Multi-label
1,1,1	Expression + Discourse + Meaning	84	Multi-label

5.5.2 Inter-rater Reliability

Inter-annotator agreement is evaluated using Cohen's kappa and raw agreement. Agreement levels range from approximately 62% to 71% across categories, while kappa values range from 0.17 to 0.23 (see Table 4). The relatively low kappa values are likely influenced by label imbalance and the subjective nature of meaning-related error annotation. These results suggest that while the task is feasible, it is inherently more challenging than surface-level error annotation.

Reliability varied across learner L1 groups and fluency levels (see Table 5 and 6). At the span level, Korean essays showed the highest agreement, with a kappa of 0.351 and observed agreement of 0.832, while Chinese essays showed the lowest span-level kappa at 0.098. Japanese essays fell between these two groups, with a span-level kappa of 0.209 and observed agreement of 0.731.

At the label level, Korean essays showed relatively stronger reliability across all three categories, especially for Expression and Discourse labels. In contrast, Chinese essays showed low kappa values for Discourse and Meaning despite moderate observed agreement, suggesting that annotators may have agreed frequently on the majority class but had difficulty consistently identifying positive cases. Japanese essays showed relatively higher agreement for Discourse and Meaning than for Expression, indicating that Expression-level judgments may have been more subjective or inconsistently interpreted for this subgroup.

Reliability also differed by fluency level. Span-level kappa was highest for B1_2 essays, followed by B1_1, while A2 and B2 essays showed lower kappa values. At the label level, B1_2 showed the strongest reliability across Expression, Discourse, and Meaning, with the highest kappa observed for Meaning. In contrast, B2 showed low or near-zero kappa values despite relatively high observed agreement for Discourse and Meaning. This pattern suggests that observed agreement alone may overestimate reliability when the label distribution is imbalanced.

Table 4. Inter-annotator reliability

Task	Kappa	Agreement

Span (error / no error)	0.17	70.1%
Expression	0.22	62.2%
Discourse	0.23	66.9%
Meaning	0.18	71.4%

Table 5. Span Level (error/no error) Reliability By L1

L1	Essays	Units	Kappa	Observed Agreement
Chinese	14	266	0.098	0.632
Japanese	5	104	0.209	0.731
Korean	6	119	**0.351**	**0.832**

Table 6. Label-level reliability by L1

L1	Label	Essays	Units	Kappa	Observed Agreement
Chinese	Expression	14	266	0.193	0.602
Chinese	Discourse	14	266	0.070	0.628
Chinese	Meaning	14	266	0.042	0.729
Japanese	Expression	5	104	0.036	0.529
Japanese	Discourse	5	104	0.302	0.692
Japanese	Meaning	5	104	0.223	0.702
Korean	Expression	6	119	0.478	0.748
Korean	Discourse	6	119	0.460	0.739
Korean	Meaning	6	119	0.299	0.689

6. Experiment

The experimental setup is designed to evaluate both the detectability and the classification of meaning-related errors. Two tasks are defined. The first task is error detection, where the model determines whether a given span contains an error. The second task is error type classification, where the model assigns one or more labels corresponding to expression, discourse, and meaning.

I use the Qwen3-8B model and evaluate it under three conditions: zero-shot prompting, few-shot prompting, and a human-designed zero-shot prompt. In the few-shot setting, a small number of labeled examples are provided in the prompt to guide the model's predictions. The exact prompts used in each condition are included in the appendix.

Performance is evaluated using precision, recall, and F1 score. For the classification task, results are reported for each label as well as macro-averaged F1.

7. Results

7.1 Task 1: Error Detection

I first evaluate the model's ability to identify whether a given span contains an error. This task is framed as a binary classification problem (error vs. no error) using the automatically extracted error spans.

Table 7: Task 1 Performance (Zero-shot)

Metric	**Score**
Accuracy	0.8957
Precision	0.9119
Recall	0.9539
F1	**0.9325**

The model achieves strong performance in the zero-shot setting, with an F1 score of 0.93. This indicates that the model is highly effective at detecting whether a span contains a potential error, even without task-specific examples.

Notably, recall is particularly high (0.95), suggesting that the model is sensitive to identifying problematic spans and rarely misses errors. Precision is also high (0.91), indicating that most predicted errors are valid. Overall, these results show that error detection is a relatively easy task for the model compared to more fine-grained classification.

Given this strong performance, I focus the remainder of the analysis on Task 2, which requires distinguishing between different types of errors.

7.2 Overall Task 2 Performance

Table 8 presents the overall performance of the model across different prompting conditions.

Table 8: Overall Task 2 Performance

Setting	Precision	Recall	F1
Zero-shot	0.5262	0.5000	**0.5128**
Few-shot	0.6490	0.4003	0.4952
Human Prompt	0.5953	0.4003	0.4787

The zero-shot setting achieves the highest overall F1 score (0.51), while the few-shot condition improves precision (0.65) but results in lower recall. This indicates that few-shot prompting makes the model more conservative, reducing over-prediction but missing more true instances.

7.3 Performance by Label Type

Table 9: Task 2 Performance by Error Type

Prompting condition	Error type	Precision	Recall	F1
Zero-shot	Expression	1.000	0.010	0.021
Zero-shot	Discourse	0.551	0.945	0.696
Zero-shot	Meaning	0.483	0.704	0.573

Few-shot	Expression	0.722	0.378	0.497
Few-shot	Discourse	0.623	0.528	0.571
Few-shot	Meaning	0.578	0.268	0.366
Human-annotator-style zero-shot	Expression	0.704	0.198	0.309
Human-annotator-style zero-shot	Discourse	0.583	0.779	0.667
Human-annotator-style zero-shot	Meaning	0.532	0.229	0.320

Table 9 shows Task 2 performance by error type across three prompting conditions. In the zero-shot setting, the model shows extremely high precision but very low recall for Expression, indicating that it rarely predicts Expression errors, but when it does, the prediction is usually correct. In contrast, Discourse and Meaning show much higher recall, especially Discourse, suggesting that the model tends to over-predict these categories.

Few-shot prompting improves Expression performance substantially, increasing F1 from 0.021 to 0.497. However, it reduces recall for Discourse and Meaning, resulting in lower F1 scores for those categories. This suggests that few-shot examples help the model recognize Expression errors but may also make the model more conservative overall.

The human-annotator-style zero-shot prompt produces more balanced results than the original zero-shot prompt for Expression, but Discourse remains the strongest category. Overall, these results show that prompt design affects the model's label preferences, especially whether it under-predicts Expression or over-predicts Discourse and Meaning.

7.4 Single vs. Multi-label Analysis

Because the annotation scheme allowed overlap among Expression, Discourse, and Meaning, spans were also analyzed based on label complexity. Single-label spans were defined as spans with exactly one positive error-type label, while multi-label spans were defined as spans with two or three positive error-type labels. In the final paired-span dataset, 162 spans were single-label cases and 228 spans were multi-label cases (see Table 10).

Table 10: Performance by Label Complexity

Setting	Type	Precision	Recall	F1
Zero-shot	Single	0.2427	0.3580	0.2893
Zero-shot	Multi	0.7919	0.5426	**0.6440**
Few-shot	Single	0.4626	0.4198	**0.4401**
Few-shot	Multi	0.8419	0.3944	0.5372
Human Prompt	Single	0.3333	0.3333	0.3333
Human Prompt	Multi	0.8255	0.4204	0.5571

In the zero-shot setting, the model performs substantially better on multi-label spans (F1 = 0.64), reflecting a tendency to over-predict multiple categories. In contrast, the few-shot condition improves performance on single-label spans (F1 = 0.44) but reduces performance on multi-label spans, suggesting that the model struggles with overlapping error types when explicit examples are provided.

7.4 Performance by L1

Table 11: Task 2 Performance by L1

Setting	L1	Precision	Recall	F1
Zero-shot	CHN	0.4775	0.4645	0.4709
Zero-shot	JPN	0.5310	0.4936	0.5116
Zero-shot	KOR	0.6265	0.5778	**0.6012**
Few-shot	CHN	0.6119	0.3661	0.4581
Few-shot	JPN	0.6383	0.3846	0.4800
Few-shot	KOR	0.7250	0.4833	**0.5800**
Human Prompt	CHN	0.5574	0.3579	0.4359
Human Prompt	JPN	0.5686	0.3718	0.4496

Human Prompt	KOR	0.6815	0.5111	**0.5841**

The model consistently performs best on Korean data across all settings. Chinese data shows the lowest recall, particularly under few-shot prompting, indicating that performance is influenced by L1-specific characteristics.

7.6 Performance by Proficiency Level

Table 12: Task 2 Performance by Fluency Level

Setting	Level	Precision	Recall	F1
Zero-shot	A2	0.6000	0.5347	**0.5654**
Zero-shot	B1_low	0.4765	0.4576	0.4669
Zero-shot	B1_high	0.5655	0.5325	0.5485
Zero-shot	B2+	0.3171	0.3824	**0.3467**
Few-shot	A2	0.6404	0.3762	0.4740
Few-shot	B1_low	0.6239	0.3842	0.4755
Few-shot	B1_high	0.7449	0.4740	**0.5794**
Few-shot	B2+	0.5417	0.3824	0.4483
Human Prompt	A2	0.6480	0.4191	**0.5090**
Human Prompt	B1_low	0.5641	0.3729	0.4490
Human Prompt	B1_high	0.6476	0.4416	0.5251
Human Prompt	B2+	0.3704	0.2941	**0.3279**

Performance is lowest for the highest proficiency group (B2+), suggesting that errors in advanced writing are more subtle and harder to detect. The best performance is observed for mid-level learners (B1 high) in the few-shot condition. However, performance by fluency is somewhat inconsistent.

7.7 Additional experiment: L1 Identification (NLI)

As an additional experiment, we evaluate whether the model can identify the writer's first language (L1) – Chinese, Japanese, or Korean – based on the annotated spans. This task serves as a way to examine whether the dataset captures meaningful L1-specific patterns and whether the model can utilize them.

I consider four prompting conditions. In the zero-shot span-only condition, the model is given only the extracted error span and is asked to predict the L1 without any additional context or examples. In the zero-shot sentence-plus-span condition, the full sentence containing the span is provided along with the marked span, allowing the model to use broader contextual information.

For few-shot prompting, I provide labeled examples in the prompt. In the few-shot span-only condition, the model receives a small number of span-level examples paired with their corresponding L1 labels, followed by a new span for prediction. In the few-shot sentence-plus-span condition, both the full sentence and the span are included in the examples and the test input, giving the model access to richer contextual cues.

Table 13 reports performance per L1 across these conditions.

Table 13: NLI Performance by L1

Condition	**L1**	**Precision**	**Recall**	**F1**
Zero-shot (span)	Chinese	0.5621	0.7974	**0.6594**
	Japanese	0.0000	0.0000	0.0000
	Korean	0.2099	0.1868	0.1977
Zero-shot (sent+span)	Chinese	0.5677	0.9604	**0.7136**
	Japanese	0.0000	0.0000	0.0000
	Korean	0.2632	0.0549	0.0909

Few-shot (span)	Chinese	0.7353	0.1126	0.1953
	Japanese	0.2514	0.5500	0.3451
	Korean	0.3631	0.7558	**0.4906**
Few-shot (sent+span)	Chinese	0.7500	0.4189	0.5376
	Japanese	0.2640	0.6500	0.3755
	Korean	0.5970	0.4651	**0.5229**

In the zero-shot settings, the model achieves high performance for Chinese (F1 = 0.66 -- 0.71), while Japanese remains at 0.00 and Korean is low (F1 = 0.09 -- 0.20). In the few-shot settings, performance for Chinese decreases (F1 = 0.20 -- 0.54), while Japanese (F1 = 0.35 -- 0.38) and Korean (F1 = 0.49 -- 0.52) improve.

8. Discussion

Based on the results in Section 7 , several consistent patterns can be observed.

First, there is a clear difference between error detection and error classification. Task 1 shows very strong performance (F1 = 0.93), while Task 2 stays around F1 ≈ 0.48–0.51. This means the model can easily tell that something is wrong, but it struggles to explain what kind of problem it is. This gap appears across all conditions and suggests that classification requires more detailed understanding than detection.

Second, performance differs across error types. Discourse errors are detected more reliably than expression and meaning errors. In contrast, meaning errors have the lowest recall (0.27), which shows that the model often fails to identify them. Expression errors improve under few-shot prompting, but still remain lower than discourse. This pattern suggests that the model relies more on visible structural cues, while errors that require interpreting meaning are harder to capture.

Third, prompting changes the model's behavior in a consistent way. The few-shot condition increases precision but decreases recall. This means the model becomes more selective and predicts fewer errors overall. A similar pattern appears in the single vs. multi-label analysis. Few-shot improves performance for single-label spans (F1 = 0.44 vs. 0.29), but reduces performance for multi-label spans (F1 = 0.54 vs. 0.64). One possible explanation is that multi-label spans often include overlapping issues, such as both structural and lexical problems. In the zero-shot setting, the model tends to assign multiple labels more freely, which increases recall. In the few-shot setting, the model follows the examples more strictly and tends to assign fewer labels, which improves precision but causes it to miss additional valid labels.

Fourth, performance varies across L1 groups. The model consistently performs best on Korean data and worst on Chinese data, especially in recall. These differences are not extremely large, but they appear across all conditions, suggesting that error patterns may differ across L1 groups and that some patterns may be easier for the model to capture than others. However, this result should be interpreted cautiously because the dataset was not fully balanced across L1 groups. Chinese essays contributed the largest number of spans, while Japanese and Korean had fewer paired spans. In addition, annotation reliability was higher for Korean than for Chinese, which may have made the Korean labels more consistent and easier for the model to learn or match. Therefore, the observed L1 differences may reflect both genuine differences in learner error patterns and differences in dataset size, label distribution, and annotation consistency.

Fifth, proficiency level also affects performance. The lowest results are observed for the highest proficiency group (B2+). One possible reason is that higher-level learners produce more grammatically correct and well-structured sentences, so there are fewer obvious errors. At the same time, the remaining errors are more likely to be subtle, such as unnatural wording or unclear meaning. These types of errors are harder to detect because they are not clearly visible on the surface. In contrast, mid-level learners (B1 high) show the best performance, especially under few-shot prompting, which suggests that their errors are more consistent and easier for the model to learn. However, this interpretation should be treated cautiously because the final paired-span dataset was not balanced across proficiency levels. The B2 group had the smallest number of essays and spans, while the A2 group had the largest number of spans. In addition, reliability was relatively low for B2, especially for Expression and Discourse labels. Therefore, the lower performance on B2 may reflect both the subtlety of higher-proficiency learner errors and limitations caused by smaller sample size and lower annotation consistency.

Finally, the NLI results show a similar pattern. In the zero-shot setting, the model is strongly biased toward predicting Chinese, which leads to high performance for that class but very low performance for the others. In the few-shot setting, performance becomes more balanced across L1 groups, but overall accuracy decreases. This suggests that zero-shot performance is influenced by bias, while few-shot prompting forces the model to make more careful distinctions. At the same time, overall performance remains moderate, which indicates that identifying L1 from error spans alone is difficult. This may be because many errors are not clearly linked to specific L1 patterns, and more detailed annotation would be needed to capture these differences.

Overall, the results suggest that the model relies heavily on surface-level patterns, struggles with overlapping error types, and may be affected by both L1 background and proficiency level.

9. Conclusion

In this study, I focused on meaning-related errors in L2 writing and proposed a span-level annotation framework with three categories: expression, discourse, and meaning. Using this dataset, I evaluated whether a large language model can detect and classify these types of errors.

The results show a clear difference between error detection and classification. While the model performs well in identifying whether a span contains an error, it struggles to distinguish between different error types. Discourse errors are detected more reliably, while meaning errors remain the most difficult. Few-shot prompting improves precision but reduces recall, and performance varies depending on label complexity, L1 background, and proficiency level.

Overall, these findings suggest that current models rely more on surface-level patterns and have difficulty handling subtle or overlapping errors. This indicates a limitation in capturing deeper aspects of language use, especially those related to meaning and interpretation.

For future work, more fine-grained annotation, such as explicitly labeling L1-related error patterns, may help improve both analysis and modeling. In addition, exploring methods that better handle multi-label and context-dependent errors could further improve performance.

10. Limitations

This study has several limitations.

First, the dataset is relatively small. Although I included multiple L1 groups and proficiency levels, the number of essays and annotated spans is limited. This may affect the stability of the results and the generalizability of the findings. A larger dataset would allow more reliable evaluation and deeper analysis.

Second, annotation reliability can be improved. While multiple annotators were used, the results suggest some inconsistency across labels, especially for meaning-related errors. One possible reason is that annotators were not trained extensively before the task. For future work, providing clearer guidelines and conducting training sessions or calibration rounds before annotation could help reduce confusion and improve consistency.

Third, the current annotation does not explicitly capture L1-specific error patterns. In particular, some expression errors may be influenced by the writer’s L1, but this is not directly labeled. Including annotators who are native speakers of each L1 group could help identify these patterns more accurately. This would not only improve the reliability of the annotation, but also make it possible to study and model L1-influenced errors more directly.

Overall, addressing these limitations would strengthen both the dataset and the analysis, and could lead to more accurate modeling of meaning-related errors in L2 writing.

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Appendices

A. GPT 4o Prompt for error span extraction

You are analyzing sentences written by English learners.

Your task is to identify spans in the sentence that may contain errors related to:
- unnatural expression (wording sounds non-native-like)
- discourse problems (structure, flow, or coherence)
- unclear meaning (difficult to understand or interpret)

Do NOT focus on minor grammatical errors unless they affect meaning or naturalness.

For the given sentence, extract only the problematic span(s).

Output format:
- Return only the span(s)
- If multiple spans exist, list them separately
- Do NOT rewrite or correct the sentence
- Do NOT provide explanations

Sentence:
{input_sentence}

B. Annotation Instruction

L2 Error Annotation Instructions
1. Purpose of the Task

In this task, you will identify **types of errors in parts of sentences** that sound **unnatural, poorly structured, or unclear** in English writing.

Your goal is to focus on:

- naturalness of expression
- sentence structure and flow
- clarity of meaning

This task is about how natural and understandable the sentence is.

Ignore grammar mistakes that do not affect naturalness or meaning.

2. How the Data is Organized

- Each **Google Sheet contains one essay**.
- Each **row corresponds to one error span** within a sentence.

- If a sentence contains **multiple errors**, the same sentence will appear in **multiple rows**, each with a different error span.
- Do **not modify the sentence text**

3. What to Do (Annotation Procedure)

1. Enter your initials in the Annotator Initial column.
2. For the **error annotation**, identify which type(s) of error the error_span represents. (Select **1** if the error span corresponds to that error type. Leave it as **0** if the error type does not apply.)
   * **Multiple labels can be marked as 1** if the error span fits more than one category.
3. In the correct_span column: Click **Agree** if you agree with the automatically generated error span. Click **Disagree** if you think the error span boundary is incorrect.

4. Error Labels

You will use the following three error labels.
A span can have **multiple labels** if needed.

Expression (Word Choice / Phrase)

Question:

Do the words sound unnatural or non-native?

Use when a phrase is understandable but sounds awkward or unnatural.

Examples:

- "give bad influence" → have a negative influence
- "make an experience" → gain experience

Reason: unnatural word or phrase choice

Discourse (Structure / Flow / Style)

Question:

Is the sentence structure, flow, or logic awkward?

Use when the sentence is structurally unnatural or poorly connected.

Examples:

- "Because smoking is bad." → incomplete clause
- "But not graduate." → sentence fragment
- "you who do many jobs" → awkward structure

Reason: structure, logic, or style problem

Meaning (Unclear / Hard to Understand)

Question:

Is the meaning unclear or difficult to understand?

Use when the intended meaning is confusing or unclear.

Examples:

- "She made herself own decision by himself" → pronoun confusion/unclear

6. Multi-label Rule

A span can have **multiple labels** if multiple problems are present.

Example:

sentence_text	expression_error	discourse_error	meaning_error
you who do excessively many jobs might pay no attention to study	☑	☑	☐

7. Annotation Examples
Error span*=the smallest continuous phrase that must be changed

A. Word choice error (Expression) → Usually one word
Example:
*pay no attention to *learn**
→ better: studying / the study
→ Label: Expression

B. Structure / Discourse error → Structure or clause is awkward
Example:
*Because *you who do excessively many jobs* might pay no attention to study*
→ better: people who have many jobs
→ Label: Discourse + Expression

C. Meaning unclear (Meaning error) → Hard to understand meaning
Example:
*He *stood on yourself own legs**
→ unclear meaning / form
Label: Meaning

C. Zero-shot prompt

Task:1
"""/no_think

You are annotating pragmatic errors in L2 English writing.

A sentence and a marked error span are given.

Task:
Decide whether the marked span is actually an error in the sentence.

Definitions:
- Expression: awkward or inappropriate wording or collocation
- Discourse: issue in structure, flow, or discourse relation
- Meaning: unclear, misleading, or pragmatically inappropriate meaning

Sentence with marked span:
{sentence_with_boundry}

Error span:
{error_span}

Rules:
- Judge only whether the marked span is an actual error in this sentence.
- Output only one number.
- 1 = yes, the marked span is an error
- 0 = no, the marked span is not an error
- Do NOT explain anything.
- Output ONLY 0 or 1.
"""

Task 2:
""/no_think

You are annotating pragmatic error types in L2 English writing.

A sentence and a marked error span are given.

Task:
Classify which pragmatic error types apply to the marked span.

Definitions:
- Expression: awkward or inappropriate wording or collocation
- Discourse: issue in structure, flow, or discourse relation
- Meaning: unclear, misleading, or pragmatically inappropriate meaning

Sentence with marked span:
{sentence_with_boundry}

```
Error span:
{error_span}

Rules:
- Judge the marked span only.
- This is a multi-label task.
- Output three binary numbers in this exact order:
expression,discourse,meaning
- Example: 1,0,1
- Do NOT explain anything.
- Output ONLY the three numbers.
"""
```

D. Few-shots Prompt on Task 2

```
"""/no_think

You are annotating pragmatic error types in L2 English writing.

A sentence and a marked error span are given.

Classify whether the marked span has each of the following labels.

Label definitions:

Expression
- The main problem is local wording.
- This includes awkward phrase choice, unnatural collocation, wrong lexical choice, or
non-native-like wording.
- The sentence is still mostly understandable.
- A small local revision would usually fix it.

Discourse
- The main problem is how ideas are connected or organized.
- This includes incorrect logical connection, broken clause relation, confusing sentence
structure across clauses, or redundancy of information.
- Do NOT use Discourse for simple grammar, spelling, punctuation, or local word choice
problems.

Meaning
- The main problem is that the intended meaning is unclear, misleading, or hard to recover.
```

- Even if local words are corrected, the message is still difficult to understand.
- Use Meaning when the reader cannot confidently infer what the writer is trying to say.

Output format:
Return exactly three binary numbers in this order:
expression,discourse,meaning

Example:
1,0,1

Important:
Use the examples below to distinguish the label boundaries carefully.
- Expression = mainly local wording or collocation problem
- Discourse = mainly connection, clause relation, structure, or redundancy of ideas
- Meaning = mainly unclear intended meaning
- Do not overuse Discourse for local grammar or phrase problems

Here are labeled contrastive examples:

{fewshot_text}

Now classify the next example.

Sentence with marked span:
{sentence_with_boundry}

Error span:
{error_span}

Output only the three binary numbers:
expression,discourse,meaning
"""

E. Human-Designed Zero-shot Prompt

"""
You are annotating error labels in L2 English writing.

4. Error Labels
You will use the following three error labels.
A span can have multiple labels if needed.

Expression (Word Choice / Phrase)

Question:
Do the words sound unnatural or non-native?
Use when a phrase is understandable but sounds awkward or unnatural.
Examples:
“give bad influence” → have a negative influence
“make an experience” → gain experience
Reason: unnatural word or phrase choice

Discourse (Structure / Flow / Style)
Question:
Is the sentence structure, flow, or logic awkward?
Use when the sentence is structurally unnatural or poorly connected.
Examples:
“Because smoking is bad.” → incomplete clause
“But not graduate.” → sentence fragment
“you who do many jobs” → awkward structure
Reason: structure, logic, or style problem

Meaning (Unclear / Hard to Understand)
Question:
Is the meaning unclear or difficult to understand?
Use when the intended meaning is confusing or unclear.
Examples:
“She made herself own decision by himself” → pronoun confusion / unclear meaning

Task:
Given a sentence and a marked error span, decide which labels apply.

Output format:
Return exactly three binary numbers in this order:
expression,discourse,meaning

Example:
1,0,1
"""